

Shiva Balaji Earla

Systems Engineer | Java Backend | GenAI & Agentic AI

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SUMMARY

Systems Engineer with 4.1+ years of experience. Expertise in building robust enterprise backend systems and distributed services using Java and Spring Boot, coupled with designing and developing GenAI and agentic AI platforms using Python. Proven ability to deliver scalable cloud solutions. Currently serving notice period; last working day 2 December 2026.

EXPERIENCE

GenAI Developer – Finance AI Platform & Forecasting Solutions

Nov 2025 — Present

Tata Consultancy Services Hyderabad, Telangana, India

- Engineered a Spec-Driven Development (SDD) workflow on GitHub Actions, accelerating implementation tasks end-to-end via autonomous subagents.
- Automated LLM evaluation CI via LangSmith, reducing PR review time by 40% through ephemeral test servers and automated sticky comments.
- Upgraded 5+ core repositories for AI readiness using Graphify, AWS Strands, and Langfuse, ensuring 100% compliance with industry-standard engineering practices.
- Architected an enterprise forecasting and data orchestration platform using Python, Darts, and Flyte across 10+ predictive data hierarchies.
- Implemented SHAP-based explainability across 4 workflows and introduced a metadata-driven, config-first architecture, cutting development and operational effort by 80%.
- Built an agentic financial analysis layer and Next.js interfaces, automating variance identification and enabling interactive data access for finance teams.

Backend Developer – Java & Distributed System

Jul 2022 — Oct 2025

Tata Consultancy Services Hyderabad, Telangana, India

- Developed Java 17 and Spring Boot backend services for an invoicing and transaction-settlement platform supporting flights, hotels, car rentals, tours, and packages for a US-based online travel platform serving 30M+ global users.
- Built and enhanced distributed backend services processing 500K+ monthly transactions, with focus on reliability, scalability, and financial data accuracy.
- Implemented real-time data streaming improvements that reduced peak processing latency by 30% across transaction-processing workflows.
- Migrated 4 legacy service monoliths to Kubernetes and introduced retry mechanisms that reduced manual processing errors by 20%.
- Led a 7-engineer delivery effort to modernize Java/Spring Boot services with zero production disruption, coordinating implementation, releases, and technical execution.
- Improved accounting pipelines and financial calculation reliability, reducing calculation errors by 20%, while strengthening observability and production support through telemetry, Splunk, PagerDuty, and incident-resolution practices.

EDUCATION

B.Tech, Electronics & Communication Engineering
Rajiv Gandhi University of Knowledge Technologies, Basar Telangana, India 8.41/10

Jun 2018 — Jun 2022

SKILLS

Java Core & Concurrency: Java 17 · Multithreading · Concurrency

Spring Ecosystem: Spring Boot · Spring Data JPA · Spring Retry · Spring AOP · Spring JDBC

Backend Frameworks & Tools: Microservices · RESTful APIs · Hibernate · JUnit · Mockito · Maven · Swagger · Lombok

Generative & Agentic AI: LLMs · Agentic AI · RAG · SHAP

AI Frameworks & Tools: LangChain · LangGraph · DeepAgents · AWS Strands · MCP · LangSmith · Langfuse · MLflow

AI Engineering & Productivity: Spec-Driven Development (SDD) · GitHub Actions · Autonomous Subagents · Superpowers · LLM Evaluation · Graphify · PyScn · mypy · uv · Uvicorn · Ruff · MegaLinter

Python & Forecasting: Python 3 · Darts · Flyte · Pytest · Time-Series Forecasting · Data Pipelines · Trino · Hive · Databricks

Cloud & Distributed Systems: AWS · GCP · Kubernetes · Docker · Kafka · Event-Driven Architecture · Cloud Migration

DevOps, CI/CD & Observability: CI/CD · Argo CD · Harness · Codefresh · Spinnaker · JFrog Nexus · SonarQube · Splunk · PagerDuty · Komodor · Observability

Web & AI Interfaces: Next.js · React · shadcn/ui · AG-UI · Google A2A · Excel Web Add-ins · JavaScript · HTML/CSS

Engineering & Collaboration: System Architecture · Technical Leadership · Agile/Scrum · GitHub · Cursor · Claude · GitHub Copilot · Context7

CERTIFICATIONS

Certificate of Merit in Hands-on AI for Real-World
IIT Kharagpur AI4ICPS I Hub Foundation

2024

AWS Certified Cloud Practitioner
Amazon Web Services

2024

Google Cloud Generative AI L400 Badges
Google Cloud

2023